

KEMEROVO NOVOSTROYKA, Russian FederationWMO: **296450**Lat: **55.250N**Lon: **86.217E**Elev: **486**StdP: **14.44**Time Zone: **7.00 (E07)**Period: **09-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.1	-30.5	-41.6	0.5	-35.7	-36.4	0.7	-30.3	14.3	22.7	12.4	18.9	0.9	140	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	85.2	65.7	82.1	64.6	79.0	63.4	68.4	79.3	67.0	77.5	65.6	75.5	5.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.8	93.5	72.3	63.3	88.7	71.0	61.9	84.4	69.9	33.1	79.6	31.9	77.2	30.8	75.8	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.6	11.9	10.4	DB	-39.4	90.6	6.3	2.8	-43.9	92.6	-47.6	94.3	-51.2	95.8	-55.8	97.8	(4)
(5)			WB	-39.5	71.5	6.3	1.2	-44.1	72.4	-47.7	73.1	-51.3	73.8	-55.9	74.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	34.6	-2.4	3.0	22.2	40.0	49.6	63.2	66.0	61.7	50.0	36.9	17.7	5.6	(6)
(7)		DBStd	26.10	16.24	14.80	10.43	8.94	8.89	7.51	5.58	5.58	7.42	9.41	14.43	17.07	(7)
(8)		HDD50	7097	1624	1315	861	316	120	7	0	1	91	416	969	1377	(8)
(9)		HDD65	11273	2089	1735	1326	749	482	119	58	133	451	870	1419	1842	(9)
(10)		CDD50	1490	0	0	0	17	108	403	495	364	92	11	0	0	(10)
(11)		CDD65	189	0	0	0	0	4	65	87	31	2	0	0	0	(11)
(12)		CDH74	2059	0	0	0	7	90	817	796	303	46	0	0	0	(12)
(13)		CDH80	502	0	0	0	1	9	222	210	51	9	0	0	0	(13)
(14)	Wind	WSAvg	4.7	4.3	4.2	5.0	5.4	6.1	4.4	3.9	3.9	4.2	4.8	5.4	5.0	(14)
(15)	Precipitation	PrecAvg	21.9	1.2	1.0	1.0	1.1	1.7	2.7	3.0	2.7	1.9	1.9	1.9	1.8	(15)
(16)		PrecMax	27.5	2.2	2.2	2.6	2.2	3.6	5.1	4.9	5.7	3.7	3.3	2.9	3.1	(16)
(17)		PrecMin	15.8	0.1	0.0	0.2	0.1	0.4	1.0	0.4	1.1	0.4	1.1	0.8	0.4	(17)
(18)		PrecStd	2.9	0.6	0.5	0.6	0.5	0.8	1.1	1.1	1.0	0.9	0.6	0.6	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.7	35.3	48.2	73.0	80.5	89.5	89.2	85.3	80.6	67.0	49.1	35.8	(19)
(20)			MCWB	31.9	31.8	39.4	54.3	58.6	65.5	67.4	67.5	61.0	53.4	43.7	33.4	(20)
(21)		2%	DB	30.3	31.6	42.6	65.6	76.6	85.4	85.3	80.9	72.3	60.4	42.1	33.5	(21)
(22)			MCWB	28.4	29.1	36.2	50.0	55.8	65.0	66.3	66.1	58.7	50.2	37.8	31.6	(22)
(23)		5%	DB	26.3	29.5	39.3	60.6	72.2	82.3	82.0	77.3	67.7	55.2	37.6	31.3	(23)
(24)			MCWB	24.6	27.5	34.3	47.4	54.5	63.8	65.9	63.9	56.0	47.4	34.6	29.5	(24)
(25)		10%	DB	21.9	24.8	36.9	55.5	66.6	79.0	78.6	73.9	63.6	50.0	34.9	27.5	(25)
(26)			MCWB	20.5	23.0	32.9	45.3	52.6	62.6	64.6	62.7	54.3	44.6	32.8	25.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.4	32.1	40.2	54.9	61.4	70.2	70.6	69.8	63.1	54.7	43.8	33.8	(27)
(28)			MCD B	33.9	34.2	46.5	68.6	74.0	81.3	82.4	80.7	75.0	63.8	47.1	35.0	(28)
(29)		2%	WB	28.3	29.5	37.0	52.2	58.5	67.5	68.8	67.3	60.1	51.4	38.7	31.9	(29)
(30)			MCD B	29.9	31.5	41.0	63.6	71.0	79.7	79.8	76.8	70.1	59.1	41.9	33.3	(30)
(31)		5%	WB	24.5	27.4	35.0	49.1	56.3	66.0	67.5	65.7	57.7	48.0	34.9	29.5	(31)
(32)			MCD B	26.2	29.7	38.7	59.1	68.5	78.2	78.0	74.5	65.9	54.8	37.1	31.2	(32)
(33)		10%	WB	20.3	23.0	32.8	45.5	53.5	64.3	66.1	64.1	55.1	44.9	33.0	25.6	(33)
(34)			MCD B	21.7	24.9	36.7	54.5	65.6	75.3	75.8	72.0	61.5	49.6	34.9	27.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.9	17.7	17.4	18.6	20.2	21.5	18.8	18.9	19.3	13.8	12.1	13.6	(35)
(36)			MCDBR	14.2	15.8	18.4	28.8	29.0	28.5	26.0	24.9	28.9	24.8	10.9	12.9	(36)
(37)			MCWBR	13.3	14.6	13.5	15.6	13.7	11.3	10.1	11.6	15.6	14.7	8.0	12.0	(37)
(38)		5% WB	MCD BR	13.7	15.6	15.6	25.8	24.3	23.4	21.1	20.9	24.5	23.4	10.4	12.1	(38)
(39)			MCWBR	13.0	14.6	12.3	15.1	13.0	10.1	9.4	10.7	13.7	14.4	8.2	11.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.297	0.333	0.382	0.409	0.413	0.411	0.437	0.414	0.378	0.362	0.333	0.303	(40)	
(41)		taud	1.997	1.949	1.888	2.022	2.190	2.288	2.216	2.261	2.304	2.207	2.015	1.946	(41)	
(42)		Ebn at Noon	182	220	239	254	263	265	254	252	246	213	164	144	(42)	
(43)		Edh at Noon	23	35	47	47	43	39	41	37	31	27	22	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	280	597	1022	1388	1597	1864	1688	1384	918	464	231	173	(44)	
(45)		RadStd	31	44	97	90	188	162	119	107	114	53	40	30	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page